TSA Section 1 - 2022

1) The correct answer is A

This is the main conclusion - other conclusions are contained within it. 'But in the future, it will be prized for its practical uses rather than for its aesthetic appeal.' This neatly sums up the main drive of the passage.

2) The correct answer is B

They doubled round 3, giving an additional 12 + 16 + 20 (as 2 * 6 = 12, add 16, add 20). This gives an additional 48 in total.

3) The correct answer is E

For communities to not be worried, it must be true that traditional shopping methods offer nothing beyond that which can be achieved online or through other novel methods.

4) The correct answer is D

Less likely to stop does not equal will not stop. Simply because a vehicle is more expensive, does not mean that it will not stop - rather that there is a theoretically lower chance of it stopping, based on this research.

5) The correct answer is E

The reasoning is:
X is only possible if Y is the case. Y must have been the case, because X is true.

6) The correct answer is B

To achieve a mixture of 6 litres with a 50% apple juice and 50% orange juice ratio, Rob will need 3 litres of each type of juice.
For Mixture 1 (40% apple, 60% orange), let the amount be x litres.
- Apple juice = 0.4 * x
- Orange juice = 0.6 * x

For Mixture 2 (70% apple, 30% orange), let the amount be y litres.
- Apple juice = 0.7 * y
- Orange juice = 0.3 * y

The total amount of each juice should be 3 litres, and the total amount of mixture should be 6 litres.
Thus, x + y = 6
And,
0.4 * x + 0.7 * y = 3 (For apple juice)
0.6 * x + 0.3 * y = 3 (For orange juice)

7) The correct answer is C

She never needs 239 or 558 - these are relevant to others, like journalists or parents, but are not used at all by lecturers.

8) The correct answer is B

B clearly demonstrates this information, with the number of independent stores higher than the number of chain stores at the end.

9) The correct answer is C

The key element here is that compensation is paid by the owner of the item that ends up hurting someone else.

10) The correct answer is D

D most likely weakens the argument the most, although it is important to note that this would not be 'unpaid housework.'

11) The correct answer is A

A is a logical conclusion. B goes too far, and E requires significant levels of assumption.

12) The correct answer is A

For normal shampoo, the daily cost is £1.96 / 40 days = £0.049 per day.
For the new brand, the cost is £1.96 - £0.36 = £1.60. The daily cost is £1.60 / 25 days = £0.064 per day.
The difference in daily cost is £0.064 - £0.049 = £0.015, or 1.5 pence per day.
So, normal shampoo is better value by 1.5 pence per day.

13) The correct answer is B

Beeches is the only cottage that satisfies the requirements. It has three bedrooms, and also has wifi, a large garden, and is within 1km of a store.

14) The correct answer is C

N is worth 3, A is worth 1

15) The correct answer is C

The main conclusion - which includes sub-conclusions - is that pollution goes beyond just the most obvious forms that we might consider. D and E, for example, are both sub-conclusions that feed into this greater conclusion.

16) The correct answer is D

For the UK to have become 'happier' we must assume that other countries have not simply become less happy.

17) The correct answer is D

The principle used is that if something is given to you for free, you can't complain if it isn't great quality. Therefore one could not complain about free legal advice - other options like E miss the mark as they are illegal rather than free, or rely on an initial outlay (A) meaning that something is not truly free.

18) The correct answer is D

Sean will save $70.
Option 1: Taking the Ferry
Ferry ticket cost: $80
Fuel cost to reach the mainland terminal: 38 miles * $1.50/mile = $57
Total cost of taking the ferry: $80 + $57 = $137
Option 2: Taking the Plane
Plane ticket cost: $50

Cost to transport car: $100
Fuel cost to reach the mainland terminal: 38 miles * $1.50/mile = $57
Total cost of taking the plane: $50 + $100 + $57 = $207
Difference between the two options: $207 - $137 = $70

19) The correct answer is B
Only Pat can satisfy the required criteria. Pat won't be on duty with Fay twice, and will have two evenings of non-consecutive work.

20) The correct answer is B
FFion's mark is incorrectly plotted.

21) The correct answer is D
The reasoning is that X is possible without doing Y. X has happened, so the individual may not have done Y.

22) The correct answer is D
The assumption being made here is that having a more diverse workforce would result in less mistakes of this type.

23) The correct answer is E
If the students simply cannot understand and reason at the level of other students, then it is unsurprising that they would not attain the same level as other students.

24) The correct answer is C
20 seconds. Let Maya's speed be x metres per second. Then Iman' s speed is 1.25 times x. The total time taken for Maya to complete the 100 metre race would be 100 divided by x seconds. Iman started 2 seconds after Maya and finished 3 seconds before her. So Iman's total time is 100 divided by x minus 5 seconds. We can set up the equation:

100 = 125 - 5 times 1.25 times x

Solve this to find x is 4 m/s, which is Maya's speed.

Now, Iman's total time for the race is 100 divided by 4 minus 5, which equals 20 seconds.

25) The correct answer is D
Dyner fish cakes work out as having the highest amount of fish per pound spent.

26) The correct answer is E
This works out as Australia (201.2) - Pakistan (3.5) which gives the difference of $197,700

27) The correct answer is B
B is the only flaw that makes sense in this context. Predicting the future based on past data is not a flaw (option C), similarly to A, E is irrelevant and presupposes a link between the two (fine and termination of franchise), and D should be

obviously incorrect.

28) The correct answer is A
This is if X, then Y. As Y is the case, X must have been the case.

29) The correct answer is B
B is the best broad conclusion as it includes other sub-conclusions within it and allows one to understand the entire passage.

30) The correct answer is C
The possibilities are 1,2 and 4 or 1, 1 and 8. There are ten possible arrangements of these digits.

31) The correct answer is E
Belle bought the same number of teas and coffees, and at least one chocolate.
1 tea and 1 coffee cost £1.50 + £2.50 = £4.00.
1 chocolate costs £3.00.
First, buy one of each: £4.00 for the tea and coffee pair and £3.00 for the chocolate. This totals £7.00, leaving £25.00.
Now, £25.00 divided by £4.00 (for each pair of tea and coffee) equals 6 sets of tea and coffee pairs.
That gives 6 teas and 6 coffees, which totals 12 drinks.
Add in the 1 chocolate and the 1 initial set of tea and coffee, and we get 12 + 2 = 14 hot drinks.

32) The correct answer is C
12 of the small cubes would satisfy these requirements.

33) The correct answer is C
The assumption made is that just because this only benefits cities, it should be suspended. 'The project will do nothing to benefit those living in smaller towns and villages in the north of the country, and so should be suspended immediately.' In other words, bringing value to cities alone is not enough. However, in reality this could be sufficient to justify the project.

34) The correct answer is D
The argument made is that rights - like freedom of speech - alone, do not mean that one can justify one's actions as being acceptable.

35) The correct answer is D
D makes sense as it is clear that if you do someone a favour, they will return it - no matter the extent to which they like you.

36) The correct answer is D
Miya would be given Come Back, as she is the child that reads the second-most. The books are ordered from the child that reads the most to the least, and come back is the second-most expensive.

37) The correct answer is C
You scored 3 chops, 4 yips, 3 creamers, and 4 glinks, totaling 47 points.
Your opponent scored 5 chops, 3 yips, 3 creamers, and 2 glinks, totaling 50 points.
The difference between you and your opponent's score is:

(5 chops - 3 chops) + (3 yips - 4 yips) + (3 creamers - 3 creamers) + (2 glinks - 4 glinks) = 2 chops - 1 yip - 2 glinks = 50 - 47 = 3 points.
We know:
A chop is worth 1 point more than a creamer.
A creamer is worth 2x the points of a glink.
Let's assume the value of a glink is 2 points:
● Then, a creamer would be worth 4 points (2x2).
● A chop would be worth 5 points (4+1).
Your score would be: 3x5(chops) + 4xY(yips) + 3x4(creamers) + 4x2(glinks) = 47
15 + 4Y + 12 + 8 = 47
4Y = 12
Y = 3 (points for a yip)
Your opponent's score would be: 5x5(chops) + 3x3(yips) + 3x4(creamers) + 2x2(glinks) = 50
25 + 9 + 12 + 4 = 50
50 = 50
Both equations match the total points. Thus, C; A glink is worth 2 points; is correct.

38) The correct answer is D
Bar 4 accurately represents the cars that were sold. Hatches make up the vast majority, closely followed by saloons - which follows the difference in number of cars from the start to end of the day seen in the pie charts.

39) The correct answer is C
The argument being made is that an action that we might perceive as equally 'moral' to another is in fact influenced by a variety of other factors, including luck.

40) The correct answer is D
Just because both sides can't be right doesn't mean that the truth is in the middle - rather, it could be that one side IS right and the other isn't.

41) The correct answer is C
Essentially this is describing social skills, so C - that social skills are not an important part of leadership - makes sense as an assumption. D and E are not stated, B is not relevant, and A is a fair assumption but would strengthen the argument.

42) The correct answer is D
55p is the only possible value that satisfies all requirements. On the first day, let's assume you start with 100p and 50p coins, making it 150p in total. You purchase two bars of chocolate, costing 2x = 110p (since we aim to find x=55p). Your change would then be 40p, likely broken down into coins of 20p and 10p to satisfy the condition that all six coins over the two days are different. On the second day, you use this 40p (20p and 10p coins) to purchase another bar of chocolate priced at 55p. However, you're 15p short. The shopkeeper could give you change as two coins of 10p and 5p, making all six coins you handle during the two days different (100p, 50p, 20p, 10p, 10p, 5p).

43) The correct answer is E
The maximum number of points that could be scored is 14. You would place the token on the square that currently has two tokens on it, at the bottom left of the grid.

44) The correct answer is D

3 & 12 is certainly a cat and mouse pair. We know that 3 is a mouse from the fifth round of the game, that saw the cards replaced. 10 must also be a mouse, making 12 a cat.

45) The correct answer is B

The reasoning is flawed, and is that because something is ostensibly visible, it shows that some opinion or fact is true - whilst it may in fact not be.

A student can enter a maximum of three events. The highest points they can earn in one event is 10 points for 1st place.

Three possible combinations of points a student can earn are:

- Three 1st places: 10 + 10 + 10 = 30 points
- Two 1st places and one 2nd place: 10 + 10 + 6 = 26 points
- One 1st place, one 2nd place, and one 3rd place: 10 + 6 + 3 = 19 points

The scores of 30, 26, and 19 can be made, as can all scores below them except for 24. Therefore, the lowest score that is not possible for a student to achieve is 24 points.

47) The correct answer is B

B is the best main conclusion as it includes other conclusions, like E, within it. D, 'Urban dwellers value community cohesion and other quality-of-life features as much as or even more than efficiency.' would also be a sub-conclusion contained within this greater conclusion.

48) The correct answer is C

You would need to pay $4, which can be calculated through breaking down the individual costs of each item from your friends' purchases.

49) The correct answer is B

The argument being made is that the BBC is establishing these figures and allowing them to earn - therefore the idea that the MPs are already established - hence their income - weakens this argument.

50) The correct answer is D

Sign 4, between Southtown and Turnborough, contains the error. The difference to Southtown is clearly incorrect when compared to Sign 3.